Inline and Display Mathematics

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\overrightarrow{\not{D}}_R \equiv e^\mu_a \gamma^a \left(\vec{\partial}_\mu + \frac{1}{2}\omega^{cd}_\mu \sigma_{cd} + ig' B_\mu \right); \quad \overleftarrow{\not{D}}_R \equiv \left(\overleftarrow{\partial}_\mu - \frac{1}{2}\omega^{cd}_\mu \sigma_{cd} - ig' B_\mu \right) \gamma^a e^\mu_a;$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\frac{d^2y}{d\tilde{\lambda}^2} - \frac{1}{2}\frac{\mathcal{W}'}{\mathcal{W}} \left(\frac{dy}{d\tilde{\lambda}}\right)^2 - \frac{1}{2}\left[\mathcal{W} - \left(\frac{dy}{d\tilde{\lambda}}\right)^2\right] \mathcal{W}^{-1}\partial_y(\mathcal{W} g_{\mu\nu}) \frac{dx^\mu}{d\tilde{\lambda}}\frac{dx^\nu}{d\tilde{\lambda}} = 0.$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$0 = K_{\omega,\mathbf{c}}(\Psi) = \partial_\lambda S_{\omega,\mathbf{c}}(\lambda\Psi)\big|_{\lambda=1} = \sum_{j=1}^{3} \langle D_j S_{\omega,\mathbf{c}}(\Psi), \psi_j\rangle = \eta \sum_{j=1}^{3} \langle D_j K_{\omega,\mathbf{c}}(\Psi), \psi_j\rangle = \eta \partial_\lambda K_{\omega,\mathbf{c}}(\lambda\Psi)\big|_{\lambda=1}.$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$I^{(1)}_\epsilon(a^2) = \frac{\pi}{2}\frac{1}{cos\frac{\pi}{2}\epsilon} \left\{ \frac{1}{a^{1+\epsilon}} + 2\sum_{n=1}^{\infty} \frac{1}{(2\pi n)^{1+\epsilon}} + 2\sum_{n=1}^{\infty} \frac{1}{(2\pi n)^{1+\epsilon}} \left[\frac{1}{(1+\frac{a^2}{4\pi^2 n^2})^{\frac{1+\epsilon}{2}}} - 1 \right] \right\} .$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\delta L_{TG} = \varepsilon^{\mu\nu\rho}(R^\alpha_{\beta\mu\nu}\delta\Gamma^\beta_{\alpha\rho} - \bar{R}^\alpha_{\beta\mu\nu}\delta\bar{\Gamma}^\beta_{\alpha\rho}) - \partial_\mu \left[\varepsilon^{\mu\nu\rho} N^\alpha_{\beta\nu}(\delta\Gamma^\beta_{\alpha\rho} + \delta\bar{\Gamma}^\beta_{\alpha\rho})\right]$$